

Segmentation of brain ventricles over MRI images

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I. INTRODUCTION

The lateral ventricles are the largest ventricular system present in the brain. The segmentation of the ventricles in medical imaging, such as MRI or CT scans, is crucial for diagnostic monitoring or research purposes.

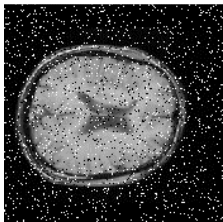


Fig. 2. MRI scan with salt & pepper noise

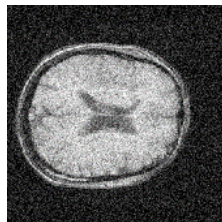


Fig. 1. MRI scan with gaussian noise

MRI scans are particularly effective for segmenting lateral ventricles, as they have a distinct shape of an X over the axial plane. However, the quality of the scan can be compromised by various factors, such as electronic noise, external interference or simply by imperfections in the MRI system. This can heavily affect the performance of the segmentation process.

In this study, we will implement a model which will perform segmentation of lateral brain ventricles and evaluate its performance by comparing the results between a clean MRI scan and an intentionally altered one by noise addition.

II. MATERIALS AND METHODS

A. Materials

The MRI dataset consisted in two .nifti1 files of different modalities. Such files contained a single MRI scan divided into 30 slices on the axial plane. Each slice had a resolution of 256x256, with voxel values ranging from 1 to 926. The first file was characterized by black ventricles on white background, while the second file presented white ventricles on black background. For simplicity, an additional pre-processing step was applied on file 2 by inverting the colors.

Most of the slices were characterized by low resolution and low contrast, therefore a pre-processing procedure

resulted to be very valuable to obtain a good performance in the segmentation part.

B. Noise introduction

Before delving into the pre-processing step, we proceeded with noise simulation. Two types of noises were studied upon, with the first one being *salt & pepper* (Fig. 2) and the second one being a *gaussian* (Fig. 1).

C. Pre processing

The pre-processing procedure consisted in three main steps: a normalization algorithm in order to ease the usage of scaling-based methods, a noise removal process to smoothen the image and finally a gamma correction to enhance the image and increase the visibility of the ventricles.

D. Noise removal

Two kinds of convolutional filters were used to remove smoothen the image and remove noise. Since the MRI scan given had a size of 256x256, filters that were too big were discarded, as they would cause a loss of resolution in image, therefore we adopted filters of size 3x3 or 5x5.

The first kind was an average based convolutional filter, which produced poor results with *salt & pepper* noise. However, it was suitable for *gaussian* noises.

The median based filter managed to hide most of the noise in the *salt & pepper* case. This is due to the function of the median, which discards binary values, particularly relevant in MRI images.

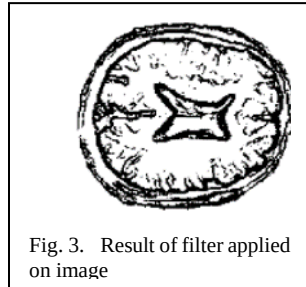
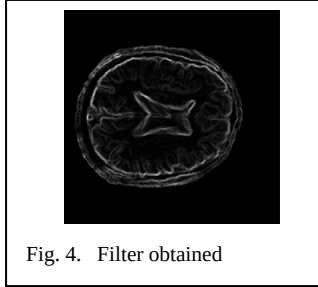
E. Gamma correction

As contrasts resulted to be quite low, gamma corrections were applied. The main goal was to darken the ventricles and lighten the background for T1, therefore the range of values was always $\gamma > 1$.

The gamma value used was mainly 1.5, however, some slices with particularly low contrast were processed by applying a gamma correction multiple times, with different parameters.

F. Segmentation

Finding the labels of the ventricles was possible thanks to edge detection. This was done by using a mask of 2 orthogonal vectors, then, filters were calculated by computing the derivatives of the matrix. This allowed us to label brain tissues and isolate the ventricles from the background.



G. Volume Computation

Once ventricle labels are extracted, each slice's volume is calculated in pixels (with depth 1) and subsequently multiplied by the pixel dimensions indicated in the header of the nifti1 file. The process is then repeated by each slice containing the ventricles, which are selected manually. The total volume is obtained by summing the singular volumes of the slices.

III. RESULTS

A. Volume

The computed volumes are shown in the table below.

	T1_16	T2_16	T1	T2
Volume (mm ³)	5846.02	7619.85	19543.22	26068.66

B. Noise

For *salt&pepper*, 3 levels of densities were applied on slice 16. At density 0.005 the labels returned were the same as the originals. Starting at density 0.01 the model would start to show a drop in performance during the filtering process as shown in Fig. 5 and Fig. 6.

For *gaussian* starting at mean=0.005, the resulting borders would turn out to be fuzzy.

C. Coefficients

Different coefficients were used to evaluate the segmentation process. The comparison was against a manually drawn ventricle label on slice T1_16. The coefficients computed are the Dice coefficient, which

measures the similarity between a manual segmentation, the sensitivity and the specificity, to assess the quantity of true positives and true negatives. Values are shown below in Table 1.

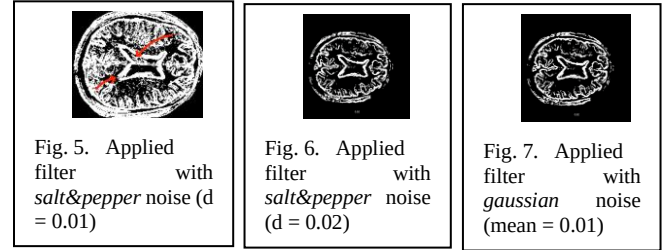


TABLE I.

	SENSITIVITY	SPECIFICITY	DICE COEFFICIENT
T1_16	0.7568	0.9748	0.8413
T2_16	0.9029	0.9176	0.8844

IV. DISCUSSION

Overall, the model performed sufficiently for both files, returning satisfactory dice coefficients.

The main obstacles faced were the variations of intensity on the ventricles, which values were very similar to the background, causing them to be hardly distinguishable by the model. When applying noise to images, the model would show a slight drop in performance but with none of its core function being compromised.

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